KG Foundations: Provenance, Maintenance, and Context Pipeline

Block 1 Day 3

Semester 1, 2026–2027 — Teaching Week 3

Session Overview

Session objective: Build trustworthy knowledge-graph reasoning pipelines using provenance, entity resolution, freshness, and context selection.
Timing target (min): 71
Topic anchors: knowledge graph; schema; instance; provenance; entity resolution; freshness; context pipeline
Padlet board: https://padlet.com/qmul/b3-d3-ap6e0bmcfc9tuhlx

Exercises

Q1 Core Concept Check

Question type Concept → Working mode Pair → Expected time 9 min

Prompt

Classify each statement as schema or instance, then rewrite one statement that is wrongly expressed. Statements: (1) Module taughtBy Lecturer, (2) EBU6505 taughtBy Dr_Riasat_Islam, (3) Room locatedIn Building, (4) Lecturer isIn Mile_End_Campus. Treat statement (4) as badly written and repair it so that it becomes a clearer graph fact or graph pattern.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 9 min
- **Deliverable:** classification + repair
- **Assessment focus:** distinguish schema structure from instance facts and detect mixed-up graph statements
- **Expected length:** 4 labels + 1 correction

How to submit

Post one pair Padlet comment with four labels and one corrected statement.

Discussion in class

Which clue tells you fastest that a statement is a pattern rather than a stored fact?

Why this helps

Useful for graph-structure questions that ask students to separate graph design from graph data.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766

Q2 Structured Problem

Question type Problem → **Working mode** Individual → **Expected time** 11 min

Prompt

Given the triples EBU6505 taughtBy Dr_Riasat_Islam, Dr_Riasat_Islam officeRoom TB_3_335, TB_3_335 locatedIn Engineering_Building, and Engineering_Building onCampus Mile_End, answer the question Where can a student go for EBU6505 lecturer support? State the graph path you use and one caveat about the answer.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 11 min
- **Deliverable:** path reasoning result
- **Assessment focus:** follow a multi-hop graph path and state a justified conclusion
- **Expected length:** 1 path + 1 answer + 1 caveat

How to submit

Post your own Padlet comment with the path, the answer, and one confidence caveat.

Discussion in class

At which step does the answer become less certain?

Why this helps

Direct practice for multi-hop reasoning and evidence-based conclusions.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766

Q3 Structured Problem

Question type Problem → **Working mode** Individual → **Expected time** 12 min

Prompt

A graph stores two room facts for the same lecturer: one came from an official timetable export yesterday, and one came from an anonymous note posted three months ago. Decide which fact should be trusted more, what provenance should be stored with it, and whether the older fact should be deleted, flagged, or kept.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** provenance judgement
- **Assessment focus:** compare evidence quality using provenance and freshness
- **Expected length:** 3 short decisions

How to submit

Post your own Padlet comment with three short decisions and one reason for each.

Discussion in class

Is a recent source always enough, or does source authority still matter?

Why this helps

Useful for questions about provenance, trust, and freshness in graph-backed agents.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766

Q4 Stretch Extension

Question type Stretch → **Working mode** Pair → **Expected time** 12 min

Prompt

You have a 600-token context budget. Create an evidence-packet strategy for a campus assistant answering: Where is today's class and who can I contact if it changes? State what must be included, what can be left out, and why.

Task details

- **Type:** stretch
- **Mode:** pair
- **Time:** 12 min
- **Deliverable:** context-budget plan
- **Assessment focus:** compress graph evidence into a small retrieval packet
- **Expected length:** 5 bullets

How to submit

Post one pair Padlet comment with five bullets covering essential evidence, optional evidence, and one item you would drop first if the budget shrinks.

Discussion in class

Which evidence item should never be removed, even under a very small budget?

Why this helps

Good extension for questions on retrieval budgets, evidence selection, and context management.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766
- doi:10.1109/TNNLS.2024.3420218
- doi:10.1016/j.iswa.2025.200541

Q5 Collaborative Mini-Case

Question type Case → **Working mode** Group → **Expected time** 15 min

Prompt

Design a weekly knowledge-graph maintenance workflow for a campus assistant. Your workflow must include ingestion, entity resolution, provenance capture, freshness checks, validation, and publication.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 15 min
- **Deliverable:** maintenance workflow
- **Assessment focus:** design a KG maintenance pipeline with trust controls
- **Expected length:** 6 steps + role labels

How to submit

Post one group Padlet comment with a numbered six-step workflow and say who or what checks each step.

Discussion in class

Which maintenance step prevents the biggest future reasoning error?

Why this helps

Supports case questions on how to keep a graph trustworthy after initial ingestion.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766

Q6 Exam-Style Constructed Response

Question type Exam → **Working mode** Pair → **Expected time** 12 min

Prompt

A student asks for the lecturer's office hour. The graph contains two conflicting facts: one from the official module page updated this morning, and one from an old departmental PDF uploaded six months ago. Explain how the agent should resolve the conflict using provenance, freshness, and source authority. Then say what the agent should do if the conflict cannot be resolved automatically.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 12 min
- **Deliverable:** structured response
- **Assessment focus:** resolve conflicting graph evidence using provenance, freshness, and source authority
- **Expected length:** 180-240 words

How to submit

Post one pair Padlet comment with a recommendation, two reasons, and one fallback action for unresolved conflict.

Discussion in class

If freshness and authority disagree, which one should dominate first?

Why this helps

Matches longer questions on evidence conflict, grounded answers, and trustworthy graph retrieval.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16, pp.701-730 and Ch.17, pp.731-766
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Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1, Q4, and Q6 are pair tasks. Q5 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which graph path gave the strongest answer, and where did uncertainty enter?
- What is the difference between keeping a graph fresh and keeping it trustworthy?
- What evidence should never be dropped from a small context packet?

Exam Transfer Note (Day-Level)

Maps to exam questions on schema versus instance, multi-hop reasoning, provenance, freshness, maintenance workflows, and evidence selection for graph-backed agents.